Table 2 (reproduction, pilot, $B = 48$ per cell). Estimated exposure effects of cumulative total exposure with model misspecification (bold cells with correct model specification).

	Estimated β (true: Cox 1, ERR 1.109)								Pairs
	True x		Naive		RRC corrected				dropped
$\rho(c, C)$	Cox	ERR	Cox	ERR	Cox.A	Cox.M	ERR.A	ERR.M	(%)
A. Mean $\hat{\beta}$ over converged fits; the paper's values in (gray), converted from its relative bias									
Cox+Additive	Cox true $\beta = 1$, additive independent error								
0.3	0.99	1.60	0.62 (0.67)	1.16 (0.80)	**0.95 (0.96)**	1.18 (0.94)	1.77 (1.17)	4.37 (1.18)	
0.6	1.07	2.03	0.70 (0.67)	1.41 (0.80)	**1.05 (0.96)**	1.24 (0.94)	2.23 (1.17)	4.16 (1.18)	
0.9	1.07	2.39	0.71 (0.69)	1.54 (0.81)	**1.07 (0.96)**	1.11 (0.95)	2.31 (1.17)	3.46 (1.17)	
Cox+Multiplicative	Cox true $\beta = 1$, multiplicative independent error								
0.3	1.01	2.98	0.71 (0.65)	1.91 (0.79)	1.13 (0.94)	**1.07 (0.94)**	3.49 (1.16)	3.03 (1.16)	
0.6	0.95	2.29	0.64 (0.66)	1.58 (0.79)	0.94 (0.94)	**0.95 (0.95)**	2.30 (1.16)	2.33 (1.15)	
0.9	1.02	2.50	0.68 (0.67)	1.65 (0.80)	1.02 (0.95)	**1.02 (0.95)**	2.48 (1.15)	2.48 (1.15)	
ERR+Additive	ERR true $\beta = 1.109$, additive independent error								
0.3	0.71	1.19	0.47 (0.58)	0.78 (0.71)	0.70 (0.85)	0.86 (0.85)	**1.16 (1.05)**	4.58 (1.04)	
0.6	0.85	1.45	0.57 (0.59)	1.10 (0.74)	0.85 (0.85)	1.00 (0.85)	**1.63 (1.05)**	2.76 (1.04)	
0.9	0.78	1.47	0.52 (0.61)	0.99 (0.76)	0.78 (0.86)	0.81 (0.85)	**1.48 (1.06)**	1.73 (1.04)	
ERR+Multiplicative	ERR true $\beta = 1.109$, multiplicative independent error								
0.3	0.59	1.38	0.48 (0.57)	1.01 (0.71)	0.75 (0.83)	0.72 (0.84)	1.68 (1.03)	**1.54 (1.04)**	
0.6	0.65	1.40	0.46 (0.58)	0.97 (0.74)	0.69 (0.84)	0.68 (0.84)	1.49 (1.04)	**1.46 (1.04)**	
0.9	0.59	1.25	0.39 (0.60)	0.83 (0.76)	0.58 (0.85)	0.59 (0.85)	1.23 (1.06)	**1.25 (1.05)**	
B. Median $\hat{\beta}$ over all fits; [number of fits not converged or stopped at the cap $\beta = 50$]									
Cox+Additive	Cox true $\beta = 1$, additive independent error								
0.3	1.02	1.55 [3]	0.61	1.04 [2]	**0.99**	1.25	1.64 [2]	4.28 [12]	–
0.6	1.09	1.78 [3]	0.73	1.18 [3]	**1.09**	1.32	1.82 [2]	3.38 [4]	–
0.9	1.18	2.11 [1]	0.78	1.42 [2]	**1.17**	1.19	2.16 [2]	2.28	–
Cox+Multiplicative	Cox true $\beta = 1$, multiplicative independent error								
0.3	1.01	2.74	0.71	1.70	1.10	**1.06**	2.66	2.62	–
0.6	0.96	1.95	0.62	1.33	0.95	**0.93**	2.03	1.93	–
0.9	1.05	2.33	0.70	1.50	1.04	**1.03**	2.20	2.25	–
ERR+Additive	ERR true $\beta = 1.109$, additive independent error								
0.3	0.73	0.98	0.46	0.73 [1]	0.69	0.88	**1.07 [1]**	2.01 [7]	–
0.6	0.84	1.22 [2]	0.56	0.82	0.83	0.99	**1.29**	1.67 [2]	–
0.9	0.73	1.22	0.50	0.74	0.73	0.77	**1.10**	1.20	–
ERR+Multiplicative	ERR true $\beta = 1.109$, multiplicative independent error								
0.3	0.61	1.33	0.54	0.91	0.77	0.76	1.38	**1.34**	–
0.6	0.63	1.20	0.45	0.85	0.68	0.68	1.38	**1.33**	–
0.9	0.59	0.97	0.38	0.66	0.56	0.57	0.98	**1.00**	–

Cox model $\lambda(t) = \lambda_0(t)\exp(\beta x(t))$ with true $\beta_{\text{Cox}} = 1$; ERR model $\lambda(t) = \lambda_0(t)(1 + \beta x(t))$ with true $\beta_{\text{ERR}} = 1.109$. Each fit is compared with the true β of the model it fits; the paper's value is $\beta(1 + \text{bias}/100)$. True x: the same risk model fitted to the true cumulative exposure, separating risk-model misspecification and small-sample bias from measurement error.
$n_1 = 2000$, $n_2 = 500$ (at most 8 pairs each), $T^* = 60$, cumulative incidence about 10%. Surrogate C has compound symmetry ($\rho = 0.5$) across years, $E(C) = 0.012$; true $c = \gamma_1 C + \epsilon$ with $\gamma_1 = 2/3$ (additive) or $\log c = \gamma_0 + \log C + \epsilon$ with $E(c) = 0.008$ (multiplicative), σ_ϵ set by $\rho(c, C)$; median x 0.21 (additive) and 0.18 (multiplicative).
Pairs dropped: share of validation pairs with $c \leq 0$ or $C \leq 0$, which the log (.M) calibration cannot use. – : no fit converged, or not recorded. A median of 50.00 means at least half of the fits stopped at the cap $\beta = 50$.